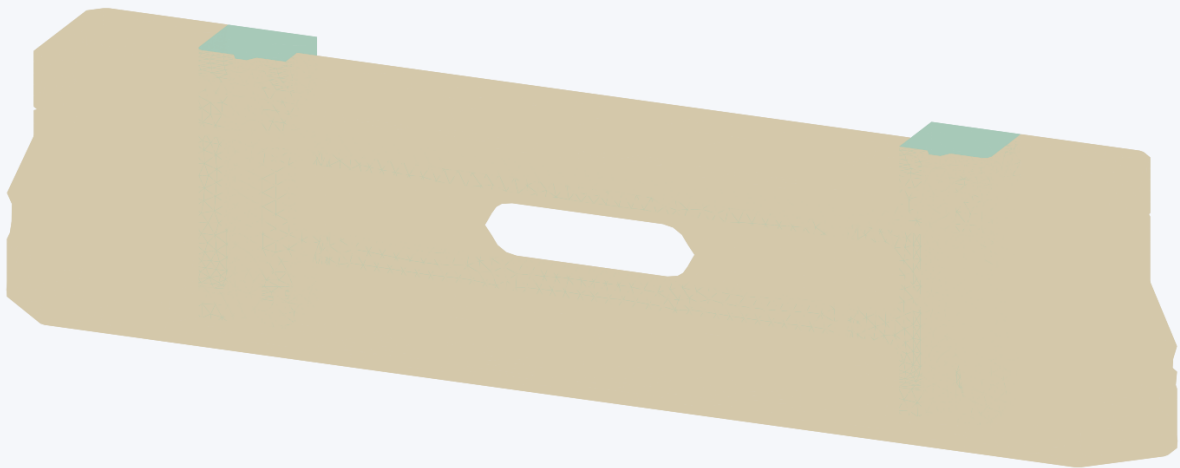


Selected-run evidence status: failed

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

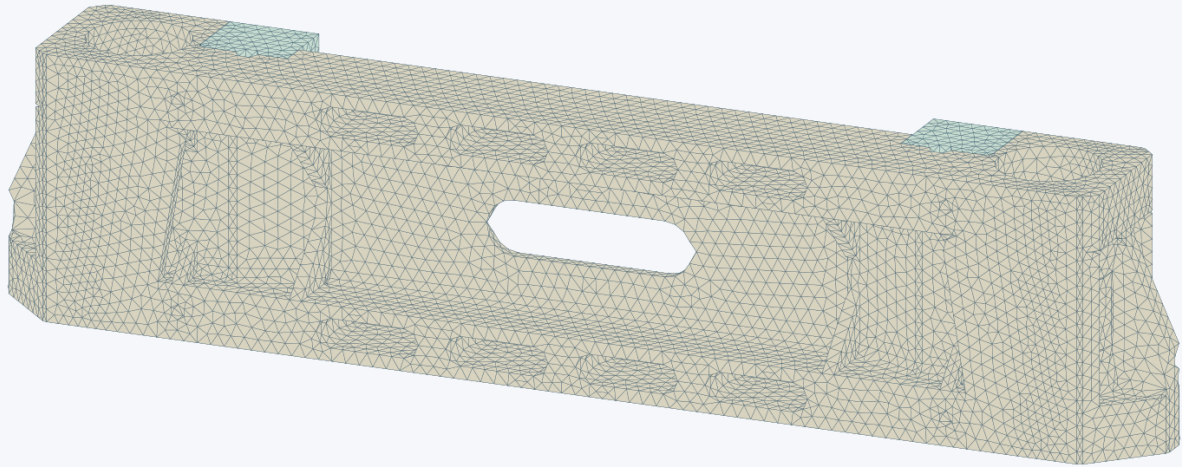
Mesh

Mesh overview and local detail

Mesh nodes	19208	
Tetrahedral elements	67240	
Meshed parts	2	
Nominal element size	4.0	mm
Minimum mean-ratio quality	0.0774	0-1

Mesh overview

Undeformed exterior mesh

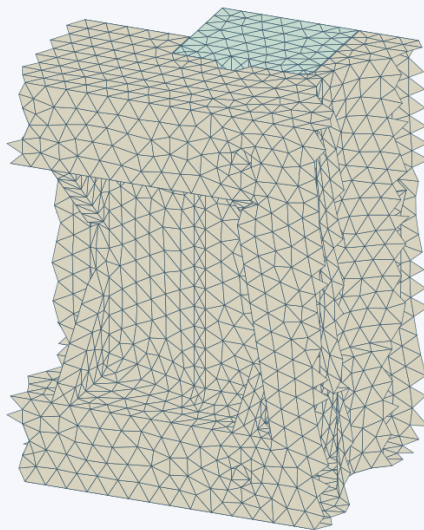


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Local mesh detail

Local mesh detail



Exterior mesh faces • undeformed geometry

Local mesh detail. Local view of the selected mesh; no local refinement claim is made.

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.04	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Motor mass	0.1616	kg
Moving mass	0.6503408	kg
Pusher mass	0.4887408	kg
Initial kinetic energy	812.926	J

Recorded material, contact and boundary assumptions

- Body and bolted cable cap flexible; four local perfectly clamped bolt collars. No bolt preload, slip or failure.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy $E=1806.6\text{MPa}$ $\nu=.35$ density 1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.
- Pusher has no external SPC constraints: all stopping reaction passes through transient contact.
- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; $HU=.05$ $BETA=0$ $DAMP=.1$ $SHAPE=1$ $FAIL=0$. Not a product material model.

- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error **MUST** be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

Solve

Recorded run status	failed
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Separate solve charts

Contact force resultant

No extracted history for this result in the selected run.

Contact force resultant. No extracted source available.

Mass-weighted forward velocity

No extracted history for this result in the selected run.

Mass-weighted forward velocity. No extracted source available.

Moving assembly travel

No extracted history for this result in the selected run.

Moving assembly travel. No extracted source available.

Foam compression

No extracted history for this result in the selected run.

Foam compression. No extracted source available.

Regional P99 von Mises stress

No extracted history for this result in the selected run.

Regional P99 von Mises stress. No extracted source available.

Global energy

No extracted history for this result in the selected run.

Global energy. No extracted source available.

Results

Run 003 - impact evidence

Evidence status: failed. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
pusher kg	0.488741
body kg	0.41625
cable cap kg	0.0724904
motors kg	0.1616
moving kg	0.650341
initial ke J	812.926
spreader each kg	0.135871

Solved mesh

Recorded mesh source: mesh_4mm_repaired. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
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Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

Numerical checks and qualification

No selected-run numerical check summary is available.

No extracted results are available for this run yet.

Images, animations and source evidence

- execution.json

- model.json

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.